TRELLO · VALIDATION SDK NODE · ITERATION 3

Aivana Node SDK Validation

I tested the official `@aivana/sdk` Node package the way an outside developer would: by checking its behavior against the Aivana docs and its README. This covers the Node part of Charter Rev2 Objective 1 / Deliverable 2 (CLI & SDK Validation). Case IDs mirror the Python report, so the two SDKs compare row by row.

SDK **@aivana/sdk 0.7.0** (npm = GitHub main) Node **25.1 (macOS) · 22 (Linux)**

Live host **dev-developers.aivana.ai** Final run **Oct 5, 2026** Status **Complete · ready for review**

69	51	16	0
test cases (43 offline, 26 live)	pass as documented	findings (3 high, 6 medium)	unexpected failures

The Node SDK has no dependencies and handles the basics well. It maps camelCase options to the API's field names, keeps `webSearch: false` and `topP: 0`, never follows redirects or sends cookies, and handles chat, `chat.continue()`, structured output and `quotas()` as documented. Because it mirrors the Python SDK, it shares most of the Python findings. It has two problems of its own in **stream cancellation**: once text starts arriving, neither an `AbortSignal` nor `timeoutMs` can stop the stream, and leaving the loop early doesn't cancel the request. Live timing also showed that **streaming starts only when the answer is nearly done**, which is API behavior that affects both SDKs.

Findings

IDs F1–F15 mean the same thing as in the Python report. N1–N3 are Node only, and P1 is a difference between the two SDKs. "API" marks behavior from the service itself, which affects both SDKs. Offline repros use a mocked `fetch`. Live repros ran on Oct 5 with the team test key.

F1 High A failed stream ends silently with a partial answer

What happens	An `error` event that arrives mid-stream is yielded like any other chunk. The README's loop (`if (chunk.event === "delta") …`) ignores it and ends normally with half an answer.
Docs say	README › Errors: "Every failure throws a subclass of `AivanaError`."
Repro	`NODE-STR-07`. Same as Python F1.
Suggest	Throw the mapped `AivanaError` subclass when an `error` event arrives.

F2 High — Out of the box, the SDK sends test keys to a host that rejects them

What happens	The default `apiBase` is `https://developers.aivana.ai`. Using the Quickstart setup exactly as written, the team's `ai_test_` key gets a 401 there. The same key works on `dev-developers.aivana.ai`.
Docs say	Quickstart: "The official SDKs already know where the Aivana API is, so there is no URL to configure."
Evidence	Confirmed live on Oct 5 (`NODE-LIV-03`). The SDK source notes the default moved back to production in 0.6.
Suggest	Sponsor decision needed: either make test keys work on the default host, or document `apiBase` for test keys and return a clearer 401.

N1 High · Node only — Once a stream starts, it can't be cancelled and never times out

What happens	The SDK detaches the caller's `signal` and clears the `timeoutMs` timer as soon as the response headers arrive. After that, aborting does nothing, and a stalled stream hangs forever.
Live evidence	Aborted at the first piece of text (4.1 s in). The stream kept going, delivered 12 more chunks and ended normally, still billing (`NODE-LIV-36`).
Docs say	README › Configuration: "Cancel an in-flight request with an AbortSignal."
Repro	`NODE-STR-08` (signal), `NODE-STR-09` (timeout), `NODE-LIV-36`
Suggest	Keep the abort and timeout wiring until the body is fully read or the generator exits, and cancel the body in the generator's `finally`.

F15 Medium · API — Streaming barely starts before the answer is done

What happens	For a ~300-word answer, the first text arrived 6.9 s into an 8.0 s request (86%). The remaining six chunks then came within about a second. A short answer showed the same pattern: first text at 2.6 s of 2.8 s.
Why it matters	The Streaming page says streaming "matters most for long answers." In practice, a streamed answer reaches the user only slightly sooner than a non-streamed one.
Repro	`NODE-LIV-37` (with timings). This is API behavior, so the Python SDK is affected too.
Suggest	Ask Aivana whether this is expected (for example, because perspectives are combined before any text is written). If so, the Streaming docs should say so and set expectations for time to first text.

N2 `Medium · Node only` Leaving a stream loop early doesn't cancel the request

What happens	`break` out of `for await` only releases the reader. The response body is never cancelled, so the server can keep generating, and charging for, the rest.
Docs say	Streaming page: stopping early should charge "the part of the answer you received."
Repro	`NODE-STR-05`
Suggest	Call `reader.cancel()` in the generator's `finally`. (Python closes the connection when the loop exits.)

F5 `Medium` Network failures and timeouts aren't AivanaErrors

What happens	A dropped connection throws fetch's `TypeError: fetch failed`. A `timeoutMs` expiry throws a bare `DOMException AbortError`, which looks the same as the caller's own abort.
Repro	`NODE-ERR-05`, `NODE-ERR-06`
Suggest	Wrap both in `AivanaError` subclasses with distinct codes, such as `network` and `timeout`.

F8 `Medium` Retrying a failed chat.send() sends the question twice

What happens	The user turn is added to the history before the request. If the request fails, it stays there, so a retry adds a duplicate.
Repro	`NODE-CHT-04`
Suggest	Add the turn to the history only after a successful response.

F11 `Medium · API` `maxTokens` isn't enforced as a ceiling

What happens	With `maxTokens: 40`, the response had 63 output tokens, all billed, with `finish_reason: "stop"`. Python saw 64–67.
Repro	`NODE-LIV-11`
Suggest	Enforce the cap, or document that it's a target rather than a hard limit.

F13 `Medium · API` Structured answers report zero output tokens

What happens	A successful `responseFormat` answer showed `usage.output_tokens: 0`, and credits covered input only. This matches Python in both of its runs.
Repro	`NODE-LIV-22`
Suggest	Ask Aivana whether this is under-metering or under-reporting.

F3 `Low` Some 422s aren't InvalidRequestError

What happens	Only allowlisted 422 codes map to `InvalidRequestError`. A real bad parameter (`temperature: 5`) was classified correctly live (`NODE-LIV-21`), so this is hardening only.
Repro	`NODE-ERR-03`

F4 `Low` Production auth errors have no requestId

What happens	The dev host includes `request_id` on a 401 and the production host doesn't. The SDK has no `X-Request-Id` header fallback.
Repro	`NODE-LIV-04` vs `NODE-LIV-02`, `NODE-ERR-04`

F6 `Low` RateLimitError drops the wait time

What happens	Neither `retry_after_ms` nor `Retry-After` is exposed. The README's own example has to back off without them.
Repro	`NODE-ERR-07`

F7 `Low` 504 isn't a retryable class

What happens	Only 502 maps to `UpstreamError`. A 504, 500 or 503 becomes the base `AivanaError`.
Repro	`NODE-ERR-08`

F14 `Low · docs` Streams send events the docs don't list

What happens	`stage` events (such as "Reviewing multiple perspectives") and a `route` event (`{model: "aivana-mmi", provider: "aivana"}`) arrive on every stream. The IP check passed: no model or provider names leaked (`NODE-LIV-35`).
Repro	`NODE-LIV-34`

P1 `Low · parity` The two SDKs send "continue" differently

What happens	Node sends `continue` inside `metadata`, while Python sends a top-level `continue` field. Both READMEs describe the feature, but they can't both match the API contract.
Repro	`NODE-GEN-12`
Suggest	Aivana should confirm which wire format the API expects and align both SDKs.

N3 `Low · docs` quotas() returns more than the README describes, and the caps are shared

What happens	`quotas()` also returns a `requests` window (a 500-requests-per-day limit), which the README doesn't mention. The day window showed 243K of 500K input tokens already used, mostly not by our runs, so the key's daily caps look shared.
Repro	`NODE-LIV-50`
Suggest	Document `requests`. For the team: check whether the test key is shared before running large test batches.

Python vs. Node

FINDING	PYTHON 0.3.0	NODE 0.7.0
F1 stream error event ignored	Yes	Yes
F2 default host rejects test keys	Yes	Yes
F3 422 allowlist	Yes	Yes
F4 no request-ID fallback	Yes	Yes
F5 network/timeout not wrapped	Yes	Yes
F6 no retry-after	Yes	Yes
F7 504 not retryable	Yes	Yes
F8 chat retry duplicates turn	Yes	Yes
F9 docs configure snippet	Yes	n/a
F10 docs/README gaps	Yes	See N3
F11 max tokens not a ceiling (API)	Yes	Yes
F12 extract shape varies	Seen once	Not seen
F13 0 output tokens on structured (API)	Yes	Yes
F14 undocumented stream events (API)	Yes	Yes
F15 streaming starts late (API)	Not measured	Yes
N1 stream can't be cancelled / no timeout	No	Yes
N2 early exit doesn't cancel	No (closes on exit)	Yes
P1 continue wire format	top-level	in metadata

Test matrix

Results from the full run on Oct 5 (macOS, Node 25.1, team test key, dev host), with a targeted rerun of NODE-LIV-30, 36 and 37 merged in. The originals are in `results/run-2026-10-05/`.

ID	AREA	CHECK	MODE	RESULT
NODE-INS-01	Install	Package installs from npm and reports its version	offline	PASS
NODE-INS-02	Install	Runs on the documented minimum (Node >= 18)	offline	PASS
NODE-INS-03	Install	Every name the docs use is exported (default + named)	offline	PASS
NODE-CFG-01	Configure	apiKey option is sent as X-API-Key	offline	PASS
NODE-CFG-05	Configure	No key: no header sent; 401 is AuthError and names the missing key	offline	PASS
NODE-CFG-06	Configure	Default apiBase is the documented hosted API; trailing slash normalised	offline	PASS
NODE-CFG-07	Configure	Docs Quickstart Node snippet runs as written (default import, intelligenceTrace)	offline	PASS
NODE-CFG-08	Configure	Requests omit cookies and never follow redirects	offline	PASS
NODE-GEN-01	Generate	Parses a real dev-API response: answer, usage, finish_reason, notices	offline	PASS
NODE-GEN-02	Generate	POSTs /v1/generate with prompt; unset options are omitted	offline	PASS
NODE-GEN-03	Generate	camelCase options reach the wire in snake_case	offline	PASS
NODE-GEN-04	Generate	System prompt > 8000 chars rejected before any request	offline	PASS
NODE-GEN-05	Generate	Attachments: mimeType normalised to mime_type	offline	PASS
NODE-GEN-06	Generate	responseFormat sent exactly as given (schema keys not re-cased); non-object rejected	offline	PASS
NODE-GEN-07	Generate	Tolerates new fields the server adds (forward compatible)	offline	PASS
NODE-GEN-08	Generate	Non-JSON 200 (proxy page) throws AivanaError, not SyntaxError	offline	PASS
NODE-GEN-09	Generate	Redirects are reported, so the key never leaves apiBase	offline	PASS
NODE-GEN-10	Generate	An AbortSignal cancels an in-flight generate() (README > Configuration)	offline	PASS
NODE-GEN-11	Generate	messages are sent instead of prompt (manual history)	offline	PASS
NODE-GEN-12	Generate	continue: true is sent as metadata.continue (README > Resuming)	offline	PASS
NODE-STR-01	Streaming	generateStream yields deltas that join into the answer; done carries usage	offline	PASS
NODE-STR-02	Streaming	Streams POST /v1/generate:stream with the same body	offline	PASS
NODE-STR-03	Streaming	responseFormat json_schema + stream refused client-side with the documented code	offline	PASS
NODE-STR-04	Streaming	HTTP 401 on the stream throws AuthError	offline	PASS

ID	AREA	CHECK	MODE	RESULT
NODE-STR-05	Streaming	Leaving a for-await loop early cancels the response body	offline	DEFECT N2
NODE-STR-06	Streaming	Unknown/trace events pass through without breaking iteration	offline	PASS
NODE-STR-07	Streaming	Mid-stream `error` event throws AivanaError ('every failure throws')	offline	DEFECT F1
NODE-STR-08	Streaming	An AbortSignal stops a stream that is already sending	offline	DEFECT N1
NODE-STR-09	Streaming	timeoutMs still applies while a stream is being read	offline	DEFECT N1
NODE-ERR-01	Errors	Each documented status maps to the documented AivanaError subclass + code	offline	PASS
NODE-ERR-02	Errors	422 details (which field is wrong) are kept on the error	offline	PASS
NODE-ERR-03	Errors	Any 422 is an InvalidRequestError (README: validation error = InvalidRequestError)	offline	DEFECT F3
NODE-ERR-04	Errors	requestId available on errors (falls back to X-Request-Id header)	offline	DEFECT F4
NODE-ERR-05	Errors	Network failure throws an AivanaError subclass ('every failure throws')	offline	DEFECT F5
NODE-ERR-06	Errors	timeoutMs expiry throws an AivanaError subclass	offline	DEFECT F5
NODE-ERR-07	Errors	RateLimitError exposes the Retry-After wait the docs say to honour	offline	DEFECT F6
NODE-ERR-08	Errors	504 (documented retryable) throws a retryable error class	offline	DEFECT F7
NODE-CHT-01	Conversations	chat() sends full history plus previous_intent / pending_action	offline	PASS
NODE-CHT-02	Conversations	chat options apply every turn; send() options win for that turn	offline	PASS
NODE-CHT-03	Conversations	reset() starts a fresh conversation	offline	PASS
NODE-CHT-04	Conversations	A failed send() does not corrupt history (retry is safe)	offline	DEFECT F8
NODE-CHT-05	Conversations	chat.continue() appends the continuation onto the last answer	offline	PASS
NODE-QTA-01	Quotas	quotas() GETs /v1/quotas with the key and returns windows	offline	PASS
NODE-LIV-01	Auth	Invalid key -> AuthError with code, status and message	live	PASS
NODE-LIV-02	Auth	Errors carry requestId (dev host)	live	PASS
NODE-LIV-03	Auth	Out of the box (default apiBase, Quickstart config) the test key works	live	DEFECT F2
NODE-LIV-04	Auth	Errors from the default (production) host also carry requestId	live	DEFECT F4
NODE-LIV-10	Generate	Quickstart call returns answer, usage, finish_reason, notices	live	PASS
NODE-LIV-11	Generate	maxTokens is a ceiling: output_tokens <= maxTokens	live	DEFECT F11
NODE-LIV-12	Generate	intelligenceTrace returns trace with steps and summary	live	PASS
NODE-LIV-13	Generate	Trace stays customer-safe: no provider/model names leak	live	PASS
NODE-LIV-14	Generate	assistantName is honoured	live	PASS
NODE-LIV-15	Generate	messages with a system message work (manual history)	live	PASS

ID	AREA	CHECK	MODE	RESULT
NODE-LIV-16	Generate	webSearch false + a question asking for the web -> web_search_off notice	live	PASS
NODE-LIV-17	Generate	stopSequences: the marker is not returned	live	PASS
NODE-LIV-18	Generate	outputShape 'extract' returns structured or structured_error	live	PASS
NODE-LIV-19	Structured	responseFormat json_schema -> validated structured (or response_format_not_available)	live	PASS
NODE-LIV-20	Structured	Unsupported schema -> InvalidRequestError invalid_response_schema	live	PASS
NODE-LIV-21	Errors	Out-of-range parameter (temperature 5) -> InvalidRequestError with details	live	PASS
NODE-LIV-22	Structured	A successful json_schema answer reports its output tokens	live	DEFECT F13
NODE-LIV-30	Streaming	Stream completes: deltas join into the answer; done carries usage/ notices	live	PASS
NODE-LIV-31	Streaming	Stream + intelligenceTrace emits trace events	live	PASS
NODE-LIV-32	Streaming	Invalid key on the stream throws AuthError	live	PASS
NODE-LIV-34	Streaming	Without intelligenceTrace, a stream sends only documented events	live	DEFECT F14
NODE-LIV-35	Streaming	Stream route/stage/trace payloads stay customer-safe	live	PASS
NODE-LIV-36	Streaming	An AbortSignal stops a live stream mid-answer	live	DEFECT N1
NODE-LIV-37	Streaming	A long answer streams incrementally (first text well before the end)	live	DEFECT F15
NODE-LIV-40	Conversations	chat() remembers earlier turns	live	PASS
NODE-LIV-50	Quotas	quotas() returns hour/day/month windows (README > Quotas)	live	PASS

Questions for Aivana

1. Should a stream be cancellable after it starts, as the README says? (N1, N2)
2. Is streaming expected to start only near the end of an answer? (F15)
3. Which host should test keys use? (F2)
4. Does the API expect `continue` at the top level or inside `metadata`? (P1)
5. Is `maxTokens` a hard cap, and are structured answers meant to report 0 output tokens? (F11, F13)
6. Is the team's test key shared? Its daily caps were already about half used. (N3)

Next steps

- Team review of severities, then move the Trello card to Done with `results/` attached.
- File the findings where Aivana confirms, using GitHub Issues on `Aivana-Inc/aivana-sdk-node` and the Python repo for shared ones.
- Add F15 to the Python report, since it's API behavior.
- Rerun when Aivana ships 0.7.1+. Fixed findings show as "FIXED? (re-check)".

Rerun it

```
cd ~/Downloads/Aivana/sdk-validation-node
bash run_validation.sh                    # offline, free
printf 'Key: ' && read -rs AIVANA_API_KEY && echo && export AIVANA_API_KEY && bash run_validation.sh
bash run_validation.sh STR                # only cases matching "STR"
```

Needs Node 18+. Live calls are spaced 6 s apart for the dev API's rate limit (about 4 minutes in total). Results: `results/matrix.md`, `matrix.csv` and `evidence.jsonl` (keys redacted).